

Agenda

- This Week
 - Class 8 – September 23rd (2:00– 3:00, 3:30 – 4:30)
 - Class 9 – September 24th (10:30 – 12:30)
 - Class 10 – September 25th (1:00 – 3:00)
- Starter Quiz
- Core Banking Platform Engineering : Processing Models (Cont'd)
- Core Banking Platform Engineering : Key Application Design Principles
 - Component Architecture
 - Platform Based
 - Software Layering
 - Multi's Support
 - Industry Standards
 - Key Stakeholders and their Interaction Enablement
- Thought Paper Presentation - ONDC (Open Network for Digital Commerce) – India (Team 6)

Note:

- Mobile Phones – Do not bring to class or Completely Switch off
- If students have any specific topic they want to discuss, share the topic and we can try to schedule
- Please be seated by your teams

Starter Quiz

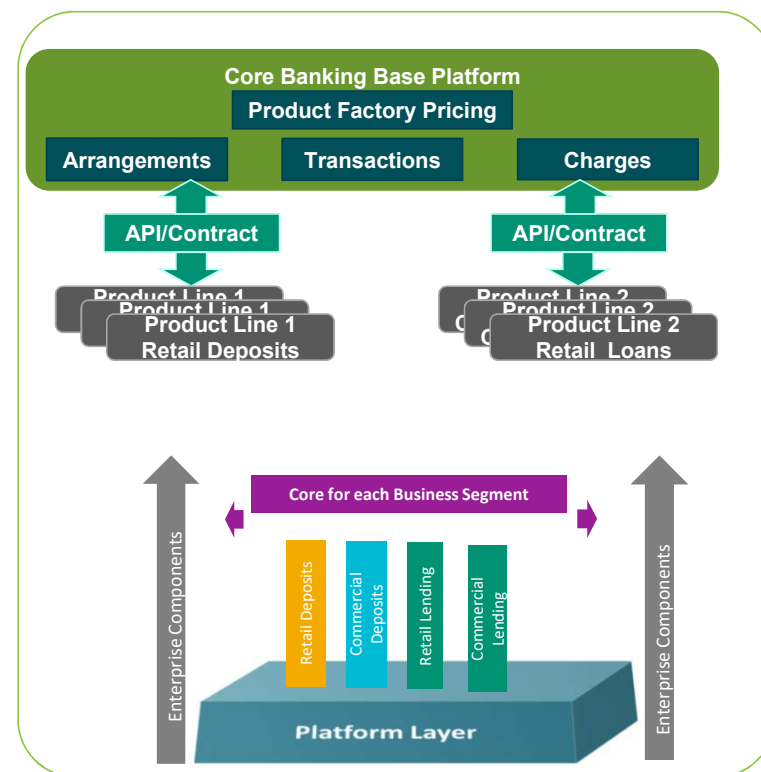
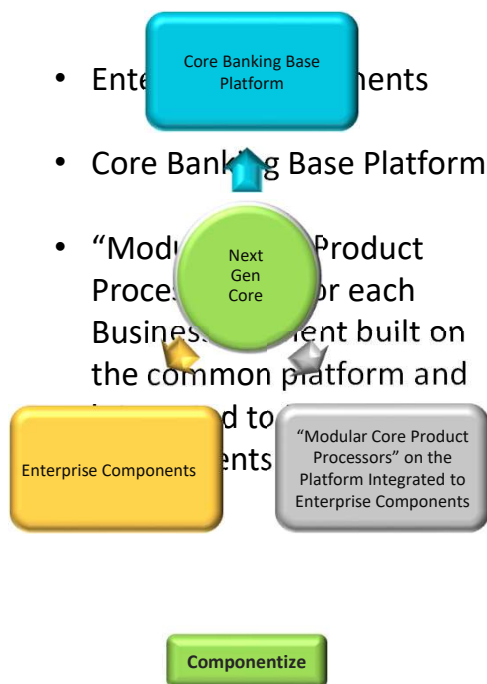
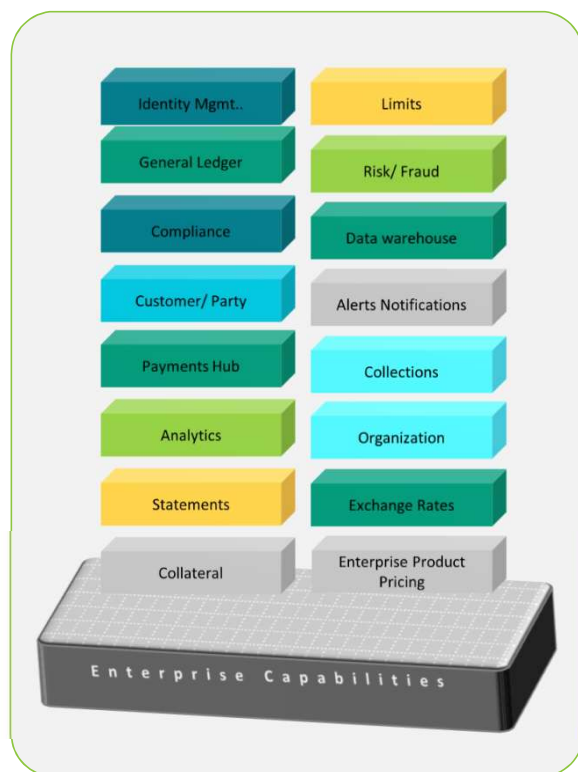
- What is the role of below Indian agency from a Crypto Exchange Perspective
 - "Financial Intelligence Unit-India (FIU-IND)"
- What is the flat tax rate on crypto gains in India as of 2025?
 - What is the Surcharge (Cess) added on the top

Core Banking Platform Engineering – Key Design Principles - Application Design

COMPONENT BASED ARCHITECTURE OVERVIEW

Component Architecture

Functional Decomposition



Modular Core Product Processors

Product Line Core – Retail Deposits - Multiple Core Product Processors

Checking and Savings



- Checking
- Savings
- MMDA

Term Deposits



- Term Deposits
 - Single Maturity Deposit
 - Renewal Deposit
 - Reinvestment Deposit
 - Step-up Deposit
 - Jumbo Deposit
- 2022 Candidates
 - Structured
 - Laddered
 - Premium
 - Discounted
 - Recurring
 - Call Deposits

Retirements



- *Traditional*
- *Roth*
- *SEP*
- *SIMPLE*
- *Keogh*

2021/2022 Candidates

- *ESA*
- *HSA*

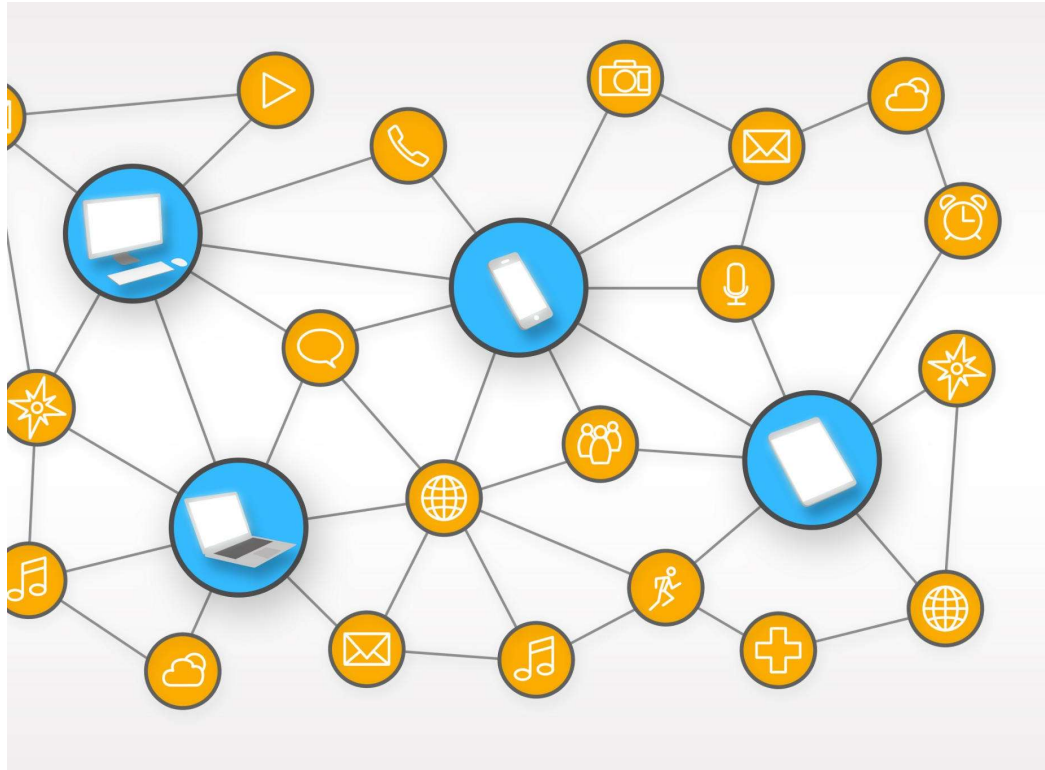
Interfaces/Apps for each Core Product Processors

Programmatic Interfaces

- API's
- Events
- Standard Extracts
- Conversion IFF

Visual Interfaces

- Configuration UI
- Servicing UI
- Reports



What is Component Based Architecture?

Modular Software Design

Component Based Architecture builds systems from independent, interchangeable modules called components.

Contrast with Monolithic Architecture

Unlike monolithic systems, CBA separates functionalities into distinct components, reducing complexity and risk.

Agility and Scalability

Component-based design enables banks to quickly adapt to market changes and scale services efficiently.

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Benefits for Core Banking

Allows Incremental legacy system replacement

Integration of third-party fintech services like payment gateways and credit scoring enhances banking agility.

Modularity, Interoperability, and Scalability

Modularity in Components

Each component manages a distinct function and can be developed independently for flexibility and efficiency.

Interoperability via APIs

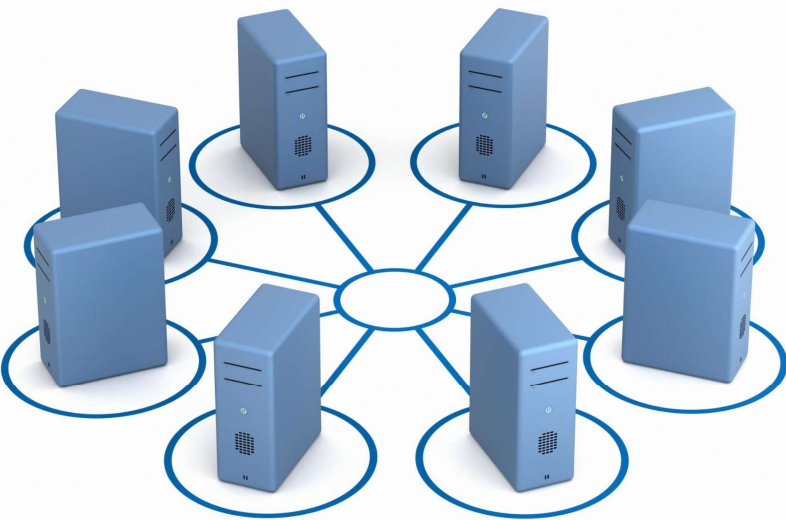
Standardized interfaces and APIs enable seamless communication between diverse components and legacy systems.

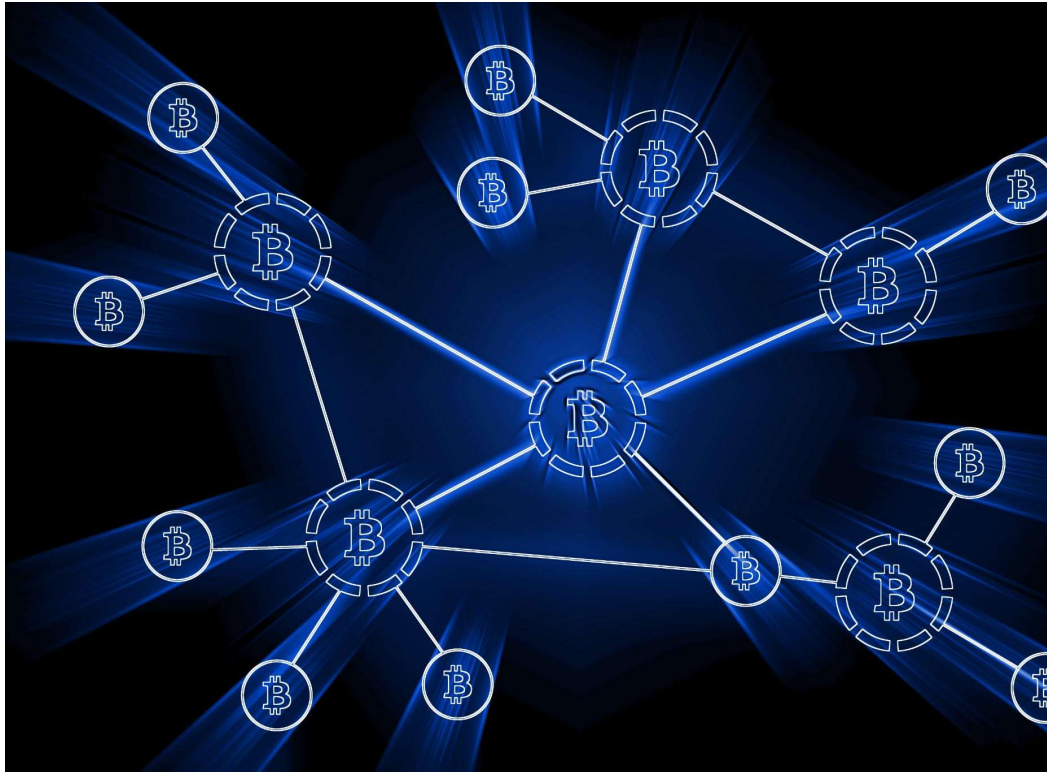
Scalability and Flexibility

Systems can add or upgrade components without disruption to handle growth and changing demands smoothly.

Reusability and Maintainability

Components can be reused across applications and isolated issues minimize downtime and operational risk.





Flexibility, Efficiency, and Innovation

Enhanced Flexibility

Component Based Architecture allows quick adaptation to regulations and technologies by updating individual parts.

Improved Efficiency

Parallel development of components reduces time-to-market and streamlines maintenance processes.

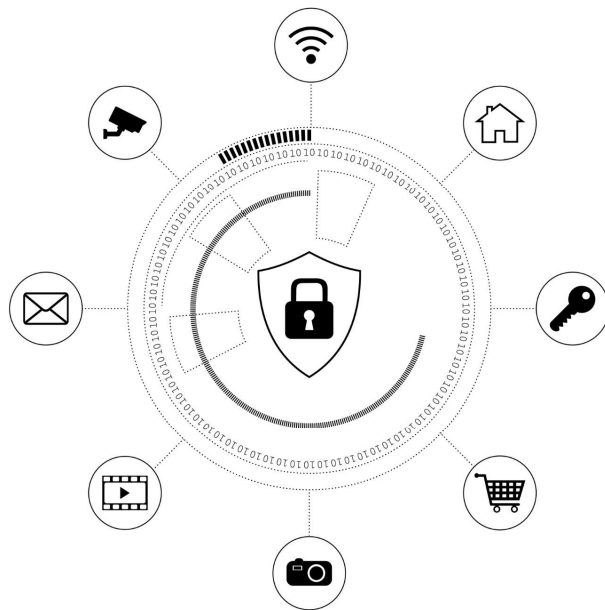
Operational Reliability

Isolated components minimize failure impact and simplify troubleshooting for banking operations.

Rapid Innovation

CBA enables integration of AI, blockchain, and open banking APIs with minimal disruption.

Integration, Security, and Change Management



Integration Complexity

Components built with different technologies must interact reliably to ensure consistent data flow and transaction integrity.

Security Challenges

Distributed architecture expands the attack surface, requiring strong authentication, authorization, and data protection.

Change Management Needs

Staff must adapt to new tools and processes, supported by disciplined governance and automated deployment pipelines.

PLATFORM BASED ARCHITECTURE OVERVIEW

Scale of Core Banking

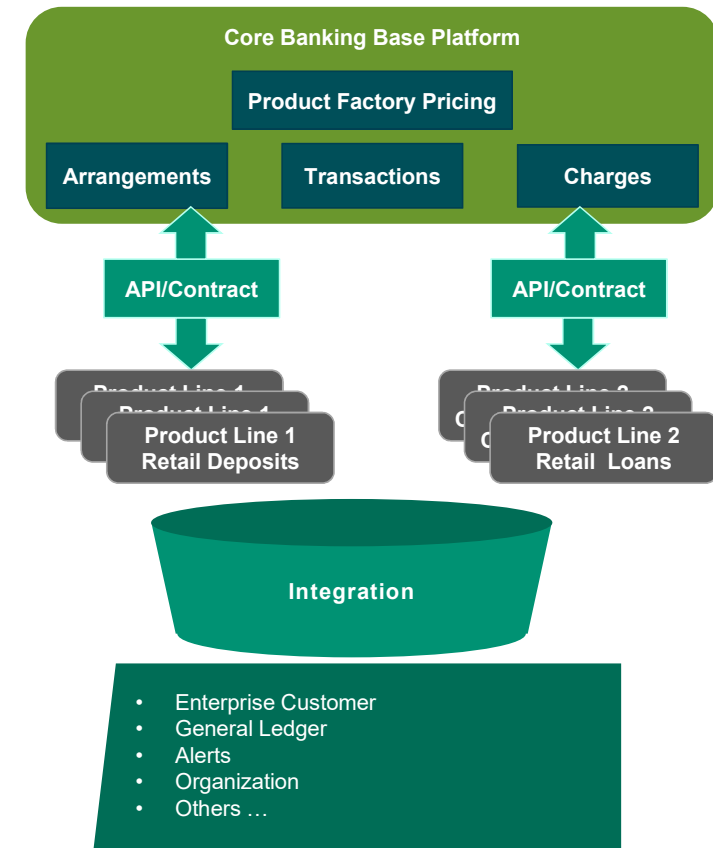
Product Lines – Product Groups – Product Type

- 4 Major Product Lines
 - Retail Deposits
 - Retail Lending
 - Commercial Deposits
 - Commercial Lending
- Each Product Line has many Product Groups e.g. Retail Deposits Product Line
 - Transactional Accounts
 - Fixed/Term Deposits
 - Retirements/Plan Accounts
- Each Product Group has many Product Types e.g. Transactional Accounts have
 - Checking/Current Account
 - Savings Account
 - Money Market Accounts
- Each Product Type can have many Products e.g. Savings Accounts can have
 - Simple Savings
 - Student Savings
 -
- Each Product Group / Product Types can have Tens of Business Processes
- Each Business Process can have a Number of Interfaces (API's, Events ...), Integrates with a Number of Components / Channel Apps, Huge Number of Business Rules etc....
- How do we Engineer, maintain and continuously Core Banking Platform covering Hundreds of Product's/Tens of Product Types ???
 - Cost, Duration, Risk
- **Platform Based Approach is a way along with Component Based Architecture**

Core Banking Base Platform

Common Account Processing Engine

- The fundamental thesis behind the utility of a common platform is that, while every “Core Product Processors” has its own unique account types, transaction types, and processes, almost all share a common set of product and pricing definition, account definition, charge processing, transaction processing and accounting patterns, making that aspect of their development and deployment ripe for componentization and reuse.
- The Common Platform will provide a substantial amount of common software engineering that is intrinsic to almost every “Core Product Line” in the financial services markets.



Architectural

- Multi Tenancy
- Multi-Currency
- Multi Language
- Multi-Platform
- Others

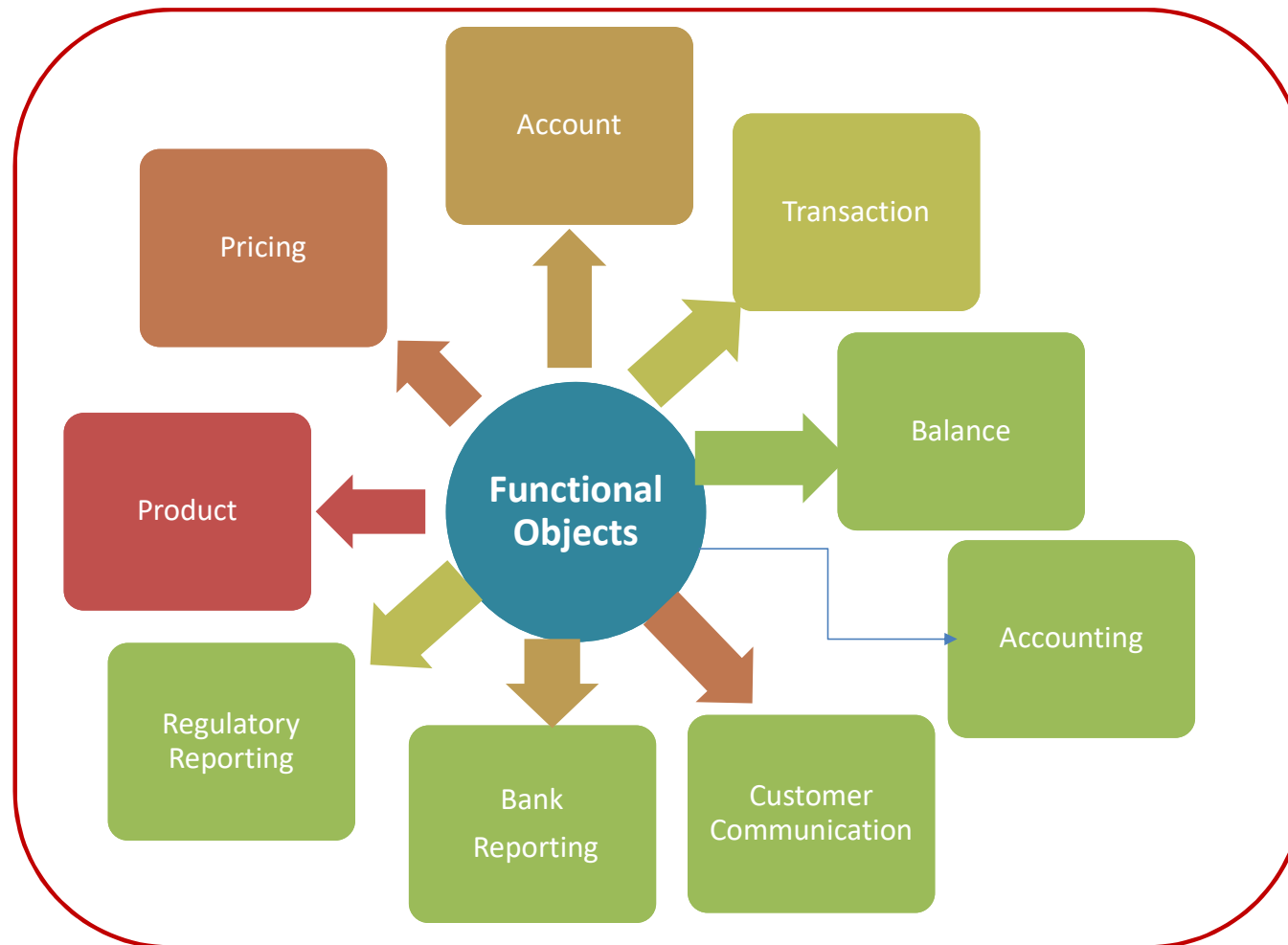
Functional

- Product Definition
- Pricing Definition
- Accounts
- Transactions,
- Balances
- ...

Integration

- Enterprise Customer
- General Ledger
- Alerts
- Organization
- Others ...

Key Generic Functional Objects in any Banking Product



Platform based “CAR” Industry Transformation

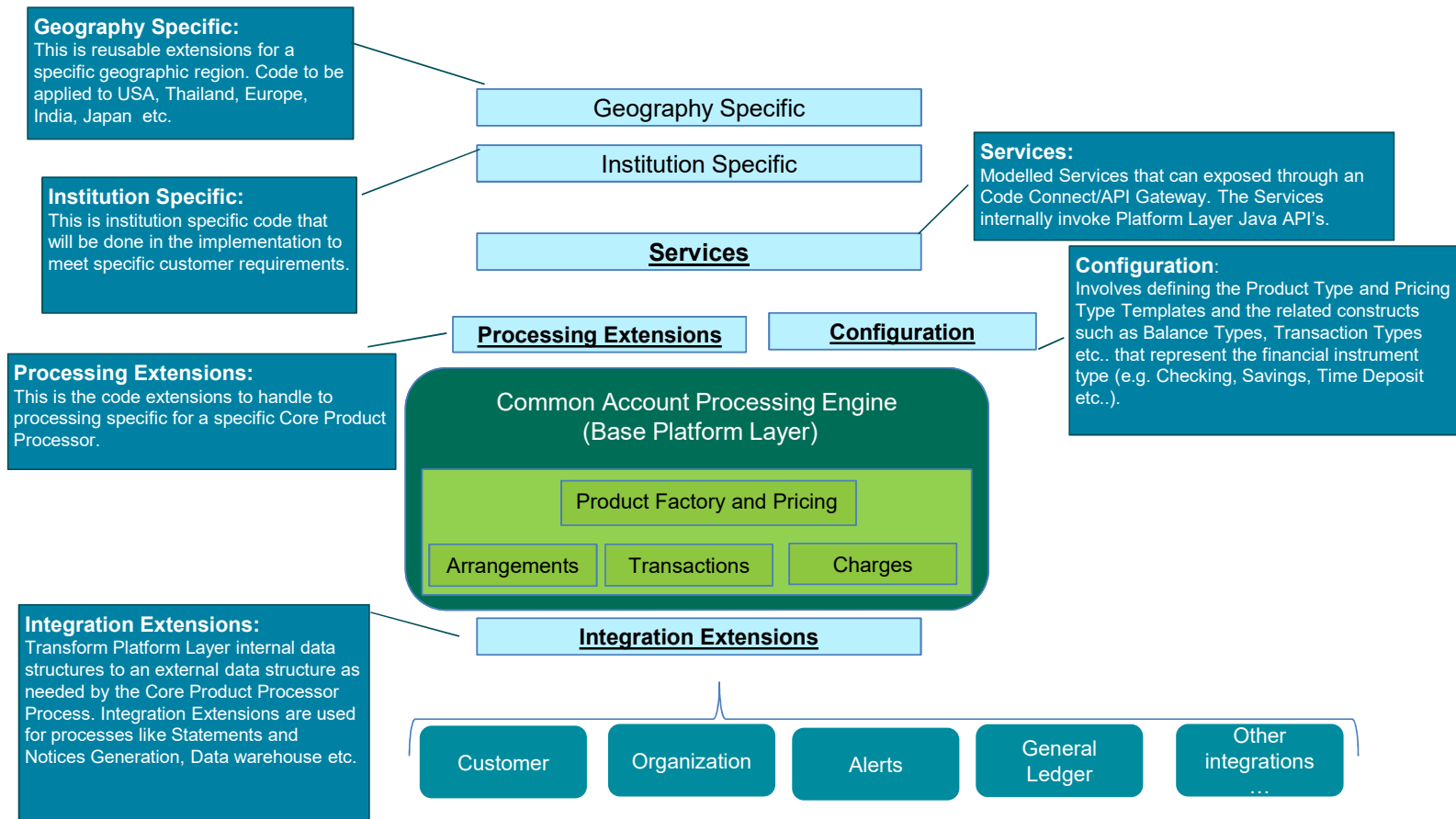
Analogous to Banking Industry “Core” Transformation with Platform based Approach



SOFTWARE LAYERING ARCHITECTURE OVERVIEW

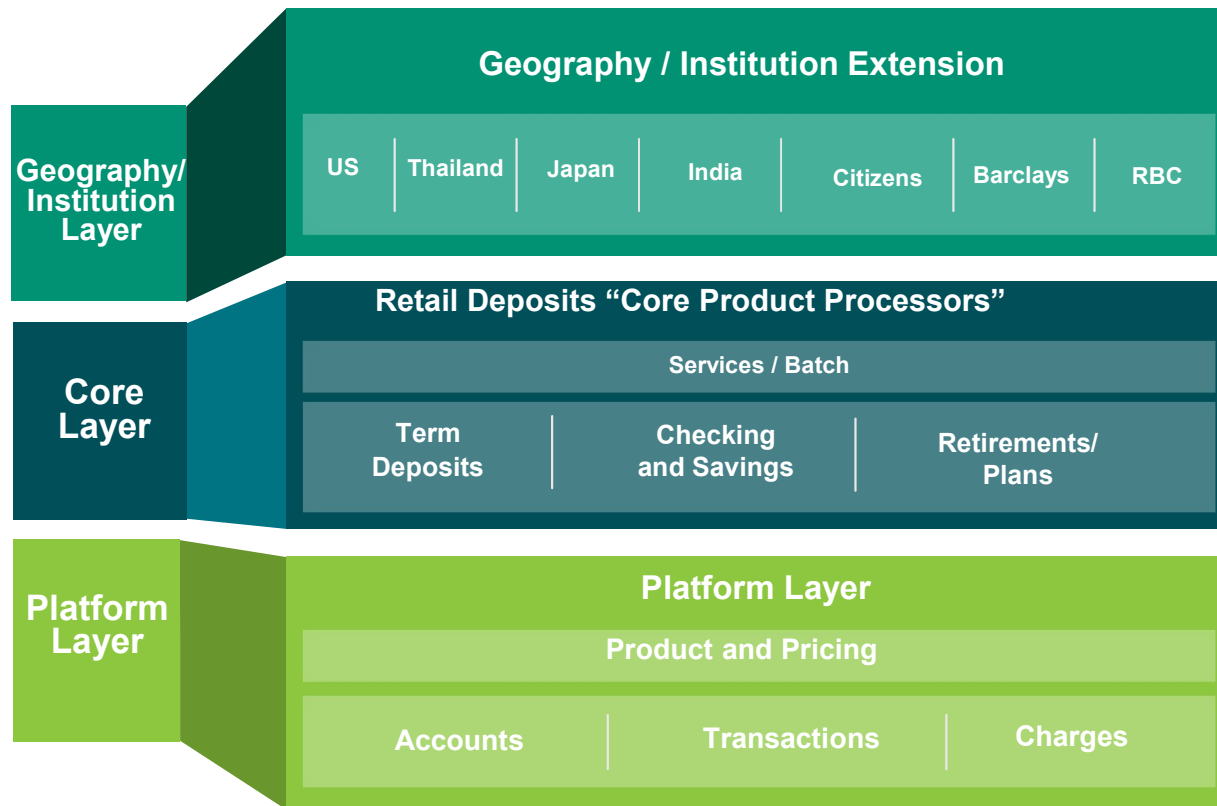
Software Layering

Core Product Processor – Built on Base Platform



Software Layering – Global Deploy ability

Built for Global Execution



Global Solution

- Easily adaptable to operate in multiple geographies
- Geography and Client Extensions to be done by different teams/organizations in parallel and evolve at their pace needed, while consuming the Base releases continuously and efficiently

MULTI'S SUPPORT

Six Multis in Core Banking

Multi-currency

Multi-currency support enables banks to handle transactions and accounts across different countries and currencies seamlessly.

Multi-language

Multi-language capability ensures core banking platforms are accessible to users from diverse linguistic backgrounds.

Multi-tenancy

Multi-tenancy lets multiple banks share application/infrastructure resources securely, supporting privacy and efficient operations within banking platforms.

Multi-time zone

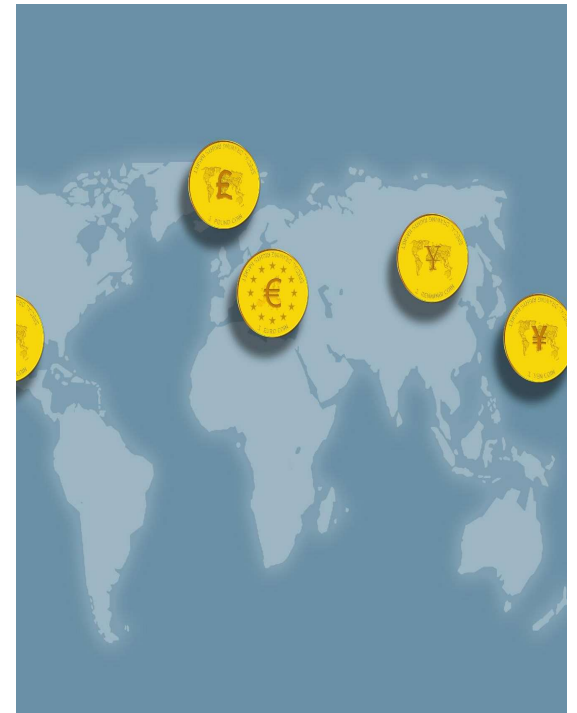
Applicable for Countries that have Multiple Time Zones like US, Russia where the time varies by time zone

Multi-Deployment Model

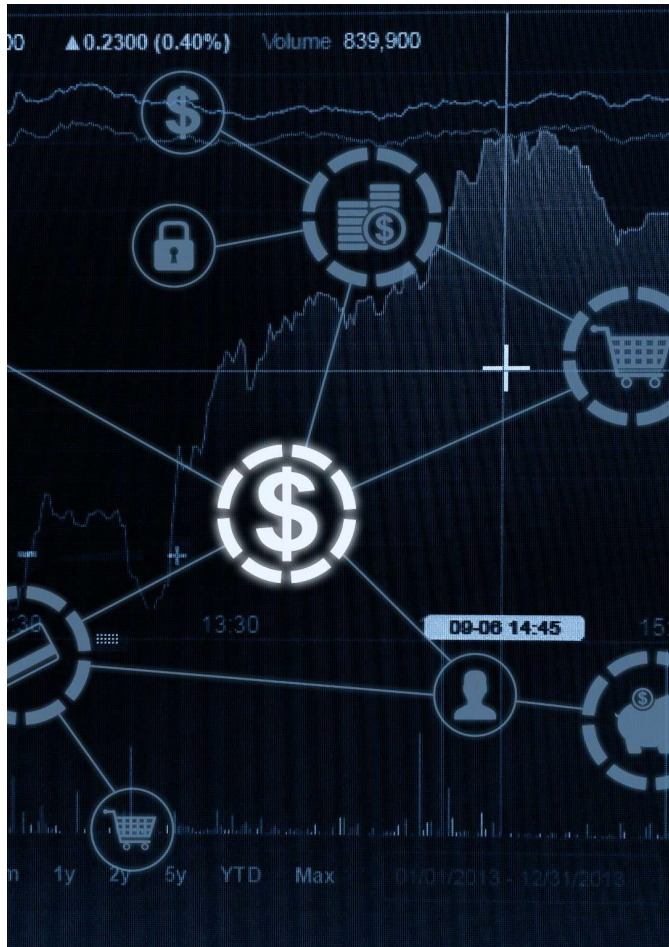
Support for On Prem/Private Cloud, Public Cloud or Hybrid Cloud

Multi- Public Cloud

Support to deploy in Multiple Cloud Providers like Azure, AWS, GCP etc.



Multi-Currency Platform Design



Support for all Currency Precisions

Support all National Currency Precisions (e.g.: 0 Decimal Yen, 2 Decimal INR/USD/EURO, 3 Decimal e.g.: Bahraini Dinar, Kuwaiti Dinar) and Cryptocurrency like Stablecoins in all Programmatic Interfaces, Visual Interfaces and Databases.

Products should be offered in multiple currencies, Accounts can be created in any of the Product offered currencies, Transactions on accounts can be done in any allowed currencies

Real-Time Multi-Currency Processing

Platforms must enable instant transactions in multiple currencies, supporting global financial operations and customer needs.

Exchange Rate Management and Conversion

Accurate exchange rate management and currency conversions (including proper rounding) ensures compliance and reliability for all currency conversions and reporting requirements.

Regulatory Reporting

Reporting by Currency and Integration with regulatory tools promote transparency and operational efficiency.

Multi-Language Platform Design

Multi-language support allows banking platforms to reach users worldwide by localizing interfaces and communications

Support Languages of Both Single Byte Character Sets (SGBS) and Double Byte Character Sets (DBCS)

Database Design – All Text Fields should be defined to support DBCS (e.g. VARCHAR(255))

Programmatic Interfaces Design – Error / Info Messages should be Language Translatable, Static Text should be encoded and Translated as per the Language in the Locale,

Visual Interfaces – Labels, Drop Downs should be displayed as per the Language in the Locale using Resource Bundles

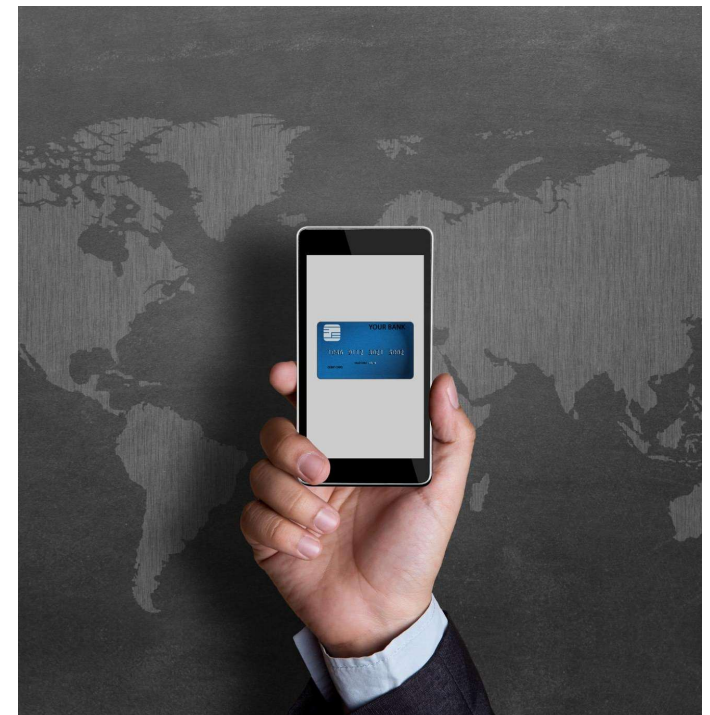
Language Text Management

Language packs provide dynamic text translation, creating seamless experiences for users in their preferred language.

Centralized management enables consistent updates and maintenance across languages, enhancing user experience and operational efficiency.

Regional Format Support

Platforms support regional formats for date, amount, ensuring consistency and accuracy for local users.



GLOBAL STANDARDS



Understanding and Adhering to Key Global Standards

Programmatic Interface Design and Documentation

- API's – Open API Specification
- Events – A synch API Specification

Payment Interfaces

- ATM/POS/Card Transactions – ISO 8583
- Other Payment Transactions – ISO 20022

Accounting Models

- GAAP
- IFRS

KEY STAKEHOLDERS AND THEIR INTERACTION ENABLEMENT



Key Stakeholders and their Interaction Enablement

Bank – Enable the Key Bank Role Activities

- Product Manager – UI and Tools to Conceptualize and Configure Products
- Servicing Staff – Front Office and Back Office/Call Centre UI's and Operational Reports
- Bank Apps Integration Engineer – Programmatic Interfaces
- Operations – Application Operational Tools and SW Promotion Pipelines
- Risk and Audit – Audits and Regulatory Reporting

Customer – Enable Customer to Transact and Communicate to Customers

- Self Service Apps
 - Mobile
 - Internet Banking
 - ATM/Kiosk
- Customer Communications
 - Alerts
 - Notices
 - Statements

Regulator – Ensure Regulatory Compliance and Share the Information to Regulator

- Reports
- Extracts
- Audits – Demonstrate Regulatory Compliance

Partners

- Payment networks and Aggregators – Programmatic Interfaces and Reconciliation Tools
- Servicing Agents – Servicing UI's, Billing Capabilities
 - Account Origination
 - Account Servicing
 - Collections